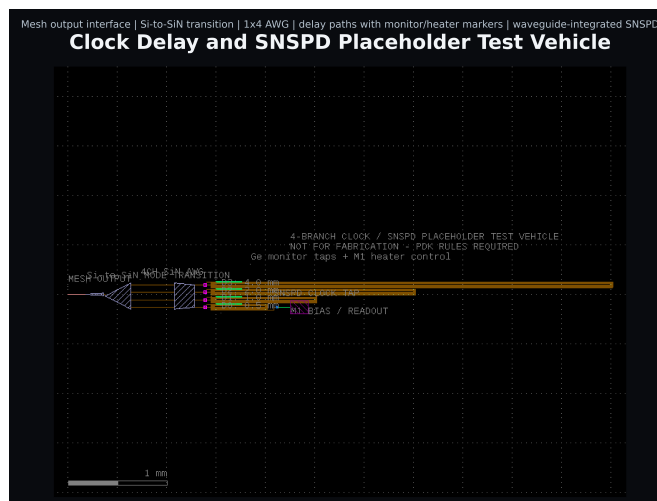


A Step-Wise Framework for Photonic-Electronic Co-Design

From napkin math to tape-out — five rungs of scrutiny for hybrid photonic-electronic wafers, illustrated with the QFlex 4-Branch Clock / SNSPD placeholder test vehicle.

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FIGURE 1



4-Branch Clock / SNSPD Placeholder Test Vehicle. Si→SiN mode transition, 4-channel SiN AWG, four SNSPD meander branches with graded lengths ($\Delta L = 1.5 / 2.0 / 1.5 / 0.5$ mm), Ge monitor taps, and M1 bias/readout pad. Provisional geometry — replace with released PDK rules before fabrication.

A WHITEPAPER

Scrutinizing an Architectural Micro-Wafer Design

A Step-Wise Framework

Introduction

The gap between a beautiful schematic and a shipping micro-wafer processor is measured not in weeks of drafting but in years of scrutiny. This is especially true for hybrid photonic–electronic architectures — designs that combine reconfigurable optical meshes, cryogenic detectors, thermo-optic control loops, and classical digital datapaths on a single die. Such systems live at the intersection of three physics regimes (coherent optical, single-photon quantum, and CMOS digital) and three design cultures (photonic circuit design, ASIC design, and systems architecture). Any one of them can kill the design, and most of the time it is not the one the architect was worried about.

What follows is a step-wise framework for scrutinizing such a design, organized from the cheapest and highest-leverage analyses to the most expensive and definitive ones. The organizing principle is simple: **the earlier a bad architecture dies, the better**, because every rung of the ladder costs roughly an order of magnitude more than the one before it. Simulation is not concept testing. Simulation is the *defense* of a concept that has already survived analytical scrutiny.

Step A — Analytical Modeling: The Napkin That Kills Bad Ideas

Before any simulator is opened, the architecture must survive a set of closed-form calculations. This stage is unglamorous, but it is where roughly 80% of unworkable designs die and should die. Four numbers dominate.

A.1 The optical loss budget.

For a mesh of depth D MZIs with per-MZI insertion loss α dB, the through-loss is $D \cdot \alpha$ dB before any coupler, routing, or grating loss is added. A 16-stage Clements mesh on silicon at $\alpha \approx 0.1$ dB/MZI is already 1.6 dB before the first splitter or fibre coupler. Add 3 dB per grating coupler, a few tenths per waveguide crossing, and the modulation depth of a Mach-Zehnder biased at quadrature, and the link budget must still leave enough optical power at the detector to exceed shot-noise-limited SNR by the margin the digital back-end requires. If it does not close on paper, no simulator will close it.

A.2 The thermal budget.

Each thermo-optic phase shifter dissipates roughly 10–30 mW for a π shift on silicon. Multiplied by a mesh of order 10^3 MZIs, the steady-state heat load can enter the tens of watts *before* the digital control fabric is powered. The question is whether the substrate, package, and cooling loop can remove that heat without introducing thermal gradients that themselves detune the mesh. This is a coupled problem, and it is often

the first physical constraint that a “just add more MZIs” architecture violates.

A.3 The energy per operation, honestly counted.

Photonic-compute papers routinely quote femtojoules per MAC by counting only the modulation energy. The honest number includes the DAC driving each heater, the TIA reading each detector, the ADC digitizing the result, the SRAM storing the operand, and the interconnect moving bits between them. Add all of it. If the total is not within an order of magnitude of a modern digital accelerator's $\sim 0.1\text{--}1$ pJ/MAC including memory movement, the architecture is not competitive on energy, regardless of what the optical core does.

A.4 The yield-adjusted throughput.

A design that requires N components each yielding at probability p produces working dice at probability p^N . A 1128-MZI mesh at 99% per-MZI yield gives $(0.99)^{1128} \approx 1\%$ working dice. This one number tells you more about commercial viability than any simulation. Either the architecture tolerates faulty MZIs (through reconfigurable routing around them), or the per-component yield target is aggressive enough that the fabrication process must be qualified specifically for it. There is no third option.

Why Step A dominates

If these four numbers do not close, the design must be revised at the schematic level — cheaply — before any further investment. This is the rung the field most often skips, and every escalation past it multiplies engineering cost by roughly $10\times$.

Step B — Behavioral / System Simulation

Once the analytics close, the next rung models each block as a transfer function or a scattering matrix and wires them together to answer *architectural* questions. Tools at this rung include Ansys Lumerical INTERCONNECT, the open-source Simphony, and modern differentiable frameworks such as SAX or photontorch. On the digital side, cycle-accurate SystemC or plain Python simulators serve.

The point is not device accuracy — it is to answer questions such as: does the log-domain normalization handoff between the optical mesh and the electronic Bayes engine actually bottleneck throughput; how much does mode-switch latency (swapping the phase-LUT bank between calculus and Bayes modes) eat into effective utilization at realistic workload mixes; and what is the sensitivity of the end-to-end error to per-stage phase noise. A behavioral model that answers these three questions is worth more than a device-level simulation of a single MMI, because it exposes *system*-level flaws that no amount of device polish can fix.

Step C — Device-Level Physical Simulation

At this rung the architect stops treating components as black boxes and confronts the electromagnetics and thermodynamics.

C.1 FDTD / EME for optical components.

Full-wave simulators (Lumerical FDTD, MEEP, Tidy3D) validate MMI splitters, grating couplers, and directional couplers. Eigenmode-expansion or beam-propagation tools handle the long, slowly-varying MZI arms. The rule is: do not simulate the whole die. Simulate the *worst-case unit cell* and confirm that its behavior matches the compact model used at Step B.

C.2 Thermal FEM for heater crosstalk.

COMSOL or Ansys thermal solves establish the crosstalk between adjacent heaters at the chosen pitch. This is often the single most important physical result for a reconfigurable-mesh architecture, because it determines whether independent-per-MZI calibration is a valid assumption or a fiction. If two adjacent heaters at your chosen pitch produce more than a few millikelvin of crosstalk into each other's waveguides, then the calibration lookup tables are coupled, and the “delta-only recalibration on mode switch” concept — which most reconfigurable schemes depend on — may not survive.

C.3 Process-variation modeling.

Wafer-scale designs are killed by variation as often as by nominal design flaws. Monte-Carlo runs over waveguide-width, sidewall-angle, and doping tolerances, driven by the foundry's actual process distributions, expose whether the design has enough margin. A mesh that works nominally but fails at $\pm 3\sigma$ on width is not a design; it is a hope.

Step D — Circuit-Level Co-Simulation

For hybrid designs this is the rung most photonic architects underinvest in. Photonic compact models (from IPKISS, Lumerical CML, the AIM Photonics PDK CMLs, or an equivalent) are imported into an analog simulator — Spectre AMS, Cadence Virtuoso, or an open equivalent — and driven by the actual

DAC, TIA, ADC, and SerDes circuits that will surround them in silicon.

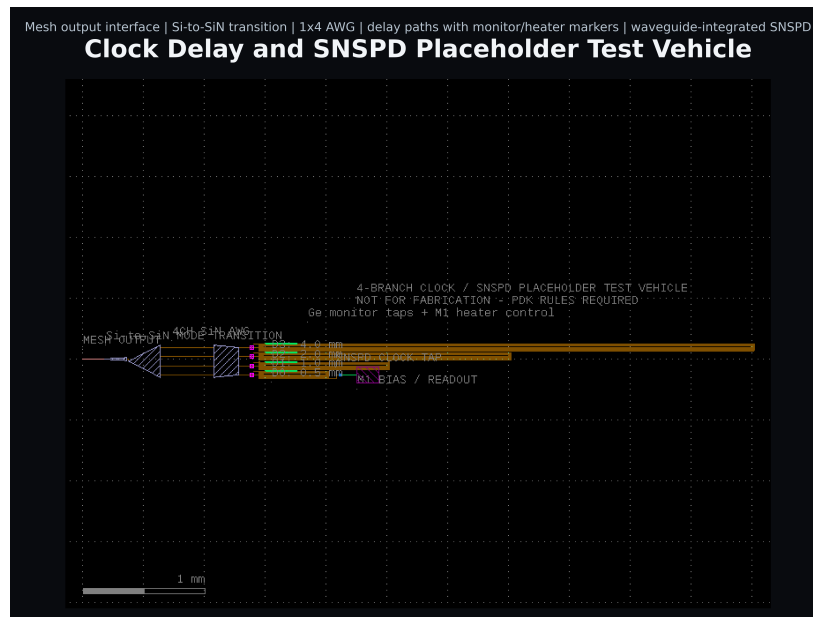
Only at this rung do you discover, for example, that the TIA input-referred noise eats the shot-noise-limited SNR margin assumed in Step A, or that the DAC settling time forces heater update rates below the throughput the architecture required. This is also where the digital and photonic teams first collide productively. Photonic architects tend to assume ideal control electronics; digital architects tend to assume ideal photonic transducers. Circuit-level co-simulation is the only stage before tape-out where those assumptions can be tested against each other.

Step E — Test-Chip Tape-Out

The only unambiguous concept test is silicon. Multi-project wafer (MPW) shuttles through AIM Photonics, imec, Europractice, or AMF make a die achievable for roughly \$10 k–\$100 k, with turnaround times of six to twelve months. Test chips at this stage should be **minimal**: not the full architecture, but a targeted set of blocks — a single mesh column, a heater-crosstalk test structure, a Ge-PD linearity ladder, a calibration-loop demonstrator. Each is designed to answer one and only one question that Steps A–D could not decisively resolve.

Everything before Step E is argument. Step E is evidence.

FIGURE 1



Worked example — a Step-E-appropriate test vehicle. The QFlex 4-Branch Clock / SNSPD placeholder isolates one question at a time: mode conversion (Si→SiN), spectral demultiplexing (4-channel SiN AWG), path-length imbalance across four graded SNSPD branches, and DC bias/readout at the M1 pad. It deliberately does not attempt the full architecture. This is what a minimal test chip should look like.

The Three Questions That Decide Competitiveness

Across all five rungs, three questions ultimately decide whether an architecture is genuinely competitive. Any design worth continuing to invest in must give sharp, defensible answers.

1. Honest energy per operation.

Count DAC, TIA, ADC, SRAM, and interconnect — not just modulation energy. If total is not within an order of magnitude of ~0.1–1 pJ/MAC including memory movement, the architecture is not competitive on energy.

2. Calibration & drift overhead.

For any reconfigurable mesh — especially one that context-switches between modes — this is the number that turns a lab demo into or away from a product. A 100 μ s recalibration on a millisecond workload boundary costs 10% throughput. A 10 ms recalibration is a death sentence.

3. Yield-adjusted throughput.

Per-component yield compounds punishingly with component count. Either the architecture tolerates faulty components through routing redundancy, or the fabrication process must be qualified aggressively enough to make the yield target credible.

Conclusion

The scrutiny of a micro-wafer architecture is not a single event but a graduated exposure of the design to progressively harsher realities: napkin math, system-level modeling, electromagnetic and thermal physics, circuit-level co-simulation, and finally silicon. Each rung filters out a class of failure modes that the rung below could not see, and each costs roughly an order of magnitude more than the one before it. The discipline of climbing the ladder in order — refusing to escalate until the current rung has been honestly satisfied — is the difference between a design that reaches fabrication and one that consumes several years of engineering effort only to fail at a stage that a spreadsheet would have caught in an afternoon.

The temptation, especially in photonic-compute research, is to skip directly from schematic to a beautiful behavioral simulation. That produces publishable plots. It does not produce competitive processors. The architectures that reach product are the ones whose designers were willing to be told, early and cheaply, that their napkin did not close — and who then went back and made it close.